

Arun Rimal

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Academic Accomplishments

- **Master of Science, Data Science** August 2024 – May 2026
Wichita State University, Fairmount, Wichita, Kansas
- **Master of Business Administration** August 2019 – 2023
DAV Business School, Lalitpur, Nepal
- **Bachelor of Electronics and Communication** August 2010 – 2014
Kantipur Engineering College, Lalitpur, Nepal

Core Competencies/Technological Stack

Programming & Scripting: Python, Node.js, TypeScript, Next.js, React, SQL, Tkinter, Selenium, Pytest. **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, PowerBI, Tableau, Jupyter. **Machine Learning & AI:** Scikit-learn, TensorFlow, Keras, PyTorch, ML Algorithms (Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest), Deep Learning Architectures (CNNs, RNNs, LSTMs, Transformers - ProsusAI/FinBERT), LLMs (OpenAI, Claude, Gemini, Langchain, Langraph). **Databases & ORM:** MSSQL, MySQL, SQLAlchemy, Prisma, Postgres, MongoDB. **Vector DBs:** ChromaDB, Pinecone. **Big Data & ETL:** PySpark, Spark, ETL pipelines, Airflow, AWS, Snowflake, Azure Data Factory. **APIs & Backend:** REST APIs, FastAPIs, GraphQL, Flask, CRUD operations. **Workflow orchestration:** LlamaIndex, Langchain, langraph. **DevOps & Containerization:** Docker.

Professional Experience

Graduate Research Assistant Jan 2025 – present

National Institute for Aviation Research (NIAR), Wichita State University

- Contributed to reverse engineering redesign initiatives by supporting component development and system enhancements using digital modeling data.

Software Engineer, Data Feb 2024 – July 2024

Infinite Computer Solutions, Hattisar, Kathmandu

- Developed and maintained ETL pipelines and stored procedures using MSSQL and SSIS for US healthcare data.

Data Engineer Sep 2021 – Feb 2024

Devfinity Nepal, Mid-Banaeshwor, Kathmandu

- Built scalable ETL pipelines using Python and SQL for API ingestion and database replication.
- Developed data warehouse pipelines in Azure Data Factory (Transact → Staging → Production).
- Developed and optimized stored procedures for data transformation and workflow automation.
- Automated web data ingestion using Python (Selenium) and integrated with Azure Blob Storage.
- Processed JSON/XML data using concurrent processing to improve performance.

DT Engineer Aug 2015 – Aug 2021

Huawei Technologies Nepal, Pulchowk, Lalitpur

- Conducted 2G/3G cluster drive testing, analyzed network KPIs, generated performance reports, ensured KPI compliance, and resolved customer network issues.

Projects

- **Financial Sentiment Analysis (ProsusAI/FinBERT):** Fine-tuned a transformer-based FinBERT model for financial sentiment analysis, achieving 87% accuracy and outperforming baseline models including original FinBERT, DistilBERT, LSTM, and Naive Bayes.
- **Credit Risk Modeling:** Built loan default prediction models using XGBoost, Logistic Regression, and Random Forest. All performed competitively, with XGBoost achieving up to 99% accuracy.
- **AWS ETL Pipeline for CSV and API Data:** Developed an automated AWS ETL pipeline to extract and validate data from CSV files and APIs, triggered via Azure SQS.

- Full-Stack AI RAG Application: Developed a secure, multi-user PDF Q&A platform using FastAPI, Next.js, and LlamaIndex, leveraging Google Gemini for reasoning and Pinecone for vector storage to enable real-time, context-aware document analysis with isolated user sessions.
- AI-Powered Tableau Dashboard Generator (In Progress): Creating an end-to-end pipeline to analyze data, recommend KPIs/visuals, and generate publication-ready Tableau dashboards.